Eli's Multiplication Facts!

Name: _____ Date: _____

Strategies that make the hard ones easier. Facts 0–12.

How to use this packet

Learn a strategy → practice a few → play a game. Short daily practice (5–10 min) beats long worksh

2s — Doubles & skip counting

2 × n means double n (or skip-count by 2s).

Example: 2 × 7 = 7 + 7 = 14. Count: 2, 4, 6, 8, 10, 12, 14.

5s — Half of a 10s fact

5 × n is half of 10 × n. Products end in 0 or 5.

Example: 5 × 8 → first 10 × 8 = 80, half is 40. Or count: 5, 10, 15, 20, 25, 30, 35, 40.

10s — Just add a zero

10 × n = n with a zero stuck on the end.

Example: 10 × 6 = 60. 10 × 12 = 120.

9s — Finger trick & the nines pattern

Finger trick: Hold up 10 fingers. Fold down finger number n. Left = tens, right = ones.

Example: 9 × 7 → fold finger 7. Left has 6, right has 3 → 63. Pattern: digits always add to 9 (6+3=9).

Doubles — square facts

n × n is a double of itself. Memorize the squares: 6×6=36, 7×7=49, 8×8=64, 9×9=81, 12×12=144.

Near-double tip: 7 × 8 = 7 × 7 + 7 = 49 + 7 = 56. (Double-and-add!)

Turn-around facts (commutative)

a × b = b × a. If you know one, you know the other!

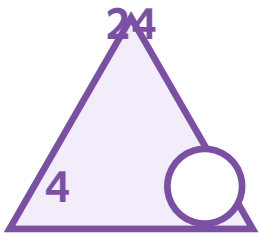
Example: 3 × 7 = 21, so 7 × 3 = 21 too. Learn the easier order first.

Hard facts to practice often: 6s, 7s, 8s, 9s, and 12s. Use the tricks above!

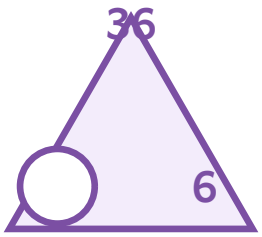
Fact Families

Name: _____ Date: _____

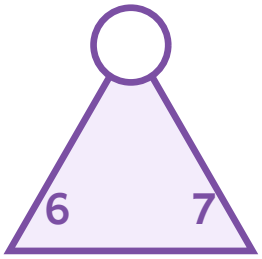
Each triangle shows related facts. Fill the blank circles. Then write the 4 sentences.



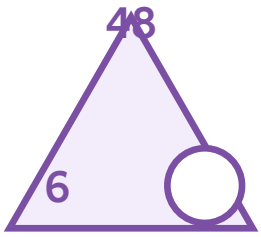
___ × ___ = _____ × ___ = ___
___ ÷ ___ = _____ ÷ ___ = ___



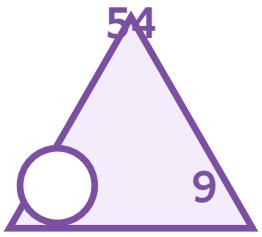
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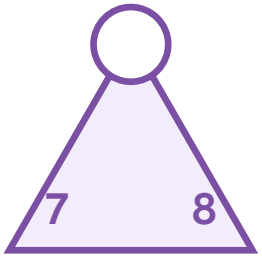
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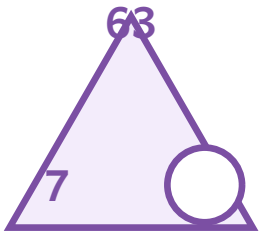
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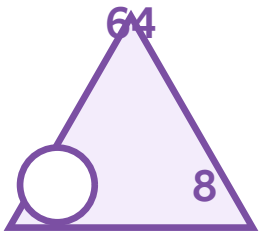
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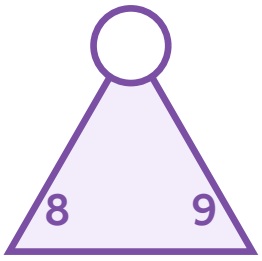
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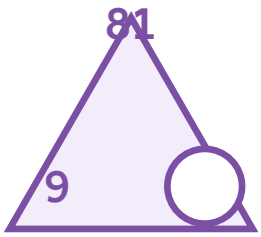
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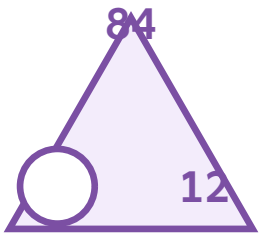
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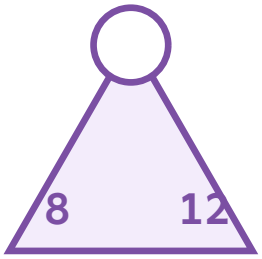
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___ × ___ = _____ × ___ = ___
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___ × ___ = _____ × ___ = ___
___ ÷ ___ = _____ ÷ ___ = ___



___ × ___ = _____ × ___ = ___
___ ÷ ___ = _____ ÷ ___ = ___

Tip: If the product is on top and the factors are the bottom corners, you can write two multiplications and two divisions from every family. Turn-around facts count!

Practice Drill A — Mixed Facts

Name: _____ Date: _____

Fill in the products. Hard facts (6s–9s, 12s) are mixed in. Take your time!

9 × 12 = _____	9 × 9 = _____	0 × 4 = _____	5 × 7 = _____
6 × 10 = _____	1 × 9 = _____	3 × 3 = _____	1 × 3 = _____
3 × 8 = _____	5 × 8 = _____	7 × 10 = _____	2 × 4 = _____
10 × 12 = _____	7 × 9 = _____	8 × 10 = _____	0 × 0 = _____
8 × 8 = _____	3 × 10 = _____	0 × 7 = _____	7 × 7 = _____
8 × 12 = _____	2 × 9 = _____	0 × 8 = _____	6 × 12 = _____
1 × 8 = _____	3 × 9 = _____	1 × 7 = _____	6 × 8 = _____
4 × 5 = _____	5 × 6 = _____	1 × 6 = _____	0 × 10 = _____
6 × 7 = _____	3 × 6 = _____	2 × 12 = _____	2 × 8 = _____
10 × 7 = _____	12 × 11 = _____	9 × 4 = _____	6 × 2 = _____

Score: _____ / 40 **Time (optional): _____ minutes**

Practice Drill B — Missing Factor

Name: _____ Date: _____

Find the missing number. Example: __ × 7 = 56 → the answer is 8.

1. 10 × _____ = 110
2. _____ × 4 = 20
3. _____ × 8 = 64
4. 12 × _____ = 144
5. 12 × _____ = 60
6. 10 × _____ = 120
7. _____ × 2 = 14
8. 12 × _____ = 24
9. 3 × _____ = 24
10. 2 × _____ = 24
11. _____ × 5 = 35
12. _____ × 10 = 120
13. _____ × 11 = 66
14. _____ × 9 = 54
15. 3 × _____ = 36
16. 10 × _____ = 30
17. 11 × _____ = 88
18. 6 × _____ = 60
19. 3 × _____ = 27
20. _____ × 12 = 84
21. 8 × _____ = 88
22. 2 × _____ = 20
23. 11 × _____ = 33
24. _____ × 11 = 77
25. _____ × 3 = 12
26. _____ × 2 = 12
27. 6 × _____ = 18
28. 9 × _____ = 72

Score: _____ / 28

Word Problems & Extra Practice

Name: _____ Date: _____

Read carefully. Show your thinking. Then finish the mini-drill.

Problem 1

Mia has 8 packs of stickers. Each pack has 7 stickers. How many stickers in all?

8 × 7 = _____ stickers

Problem 2

A soccer team practices 6 days a week for 9 weeks. How many practice days is that?

6 × 9 = _____ days

Problem 3

There are 12 muffins in a box. Mom buys 4 boxes. How many muffins?

12 × 4 = _____ muffins

Problem 4

Eli reads 9 pages every night for 8 nights. How many pages did he read?

9 × 8 = _____ pages

Problem 5

A garden has 7 rows of tomatoes with 12 plants in each row. How many tomato plants?

7 × 12 = _____ plants

Problem 6

Sam shares 56 crayons equally into 7 boxes. How many crayons in each box?

56 ÷ 7 = _____ (think: __ × 7 = 56)

Quick related facts — fill both turn-arounds:

6 × 8 = _____ and 8 × 6 = _____ 7 × 9 = _____ and 9 × 7 = _____

8 × 12 = _____ and 12 × 8 = _____ 9 × 6 = _____ and 6 × 9 = _____

Timed Challenge — Beat the Clock!

Name: _____ Date: _____

Try to finish in under 3 minutes. Write your time and score at the bottom. Ready… set… go!

Start time: _____ End time: _____ Total minutes: _____ Score: _____ / 40

1. 5 × 8 = _____	2. 5 × 11 = _____	3. 7 × 12 = _____	4. 7 × 7 = _____
5. 1 × 8 = _____	6. 4 × 12 = _____	7. 3 × 8 = _____	8. 6 × 12 = _____
9. 0 × 9 = _____	10. 1 × 7 = _____	11. 3 × 5 = _____	12. 6 × 6 = _____
13. 3 × 11 = _____	14. 5 × 12 = _____	15. 1 × 6 = _____	16. 2 × 9 = _____
17. 2 × 6 = _____	18. 4 × 10 = _____	19. 1 × 9 = _____	20. 4 × 9 = _____
21. 0 × 12 = _____	22. 3 × 7 = _____	23. 3 × 6 = _____	24. 3 × 12 = _____
25. 6 × 10 = _____	26. 2 × 10 = _____	27. 8 × 11 = _____	28. 9 × 9 = _____
29. 1 × 3 = _____	30. 4 × 8 = _____	31. 4 × 6 = _____	32. 0 × 2 = _____
33. 6 × 7 = _____	34. 0 × 7 = _____	35. 0 × 6 = _____	36. 0 × 11 = _____
37. 9 × 12 = _____	38. 0 × 3 = _____	39. 4 × 11 = _____	40. 7 × 8 = _____

How did it feel? Circle one: Easy Just right Tricky Need more practice

My records

Best time: _____ min Best score: _____ / 40 Date of best: _____

Next goal: _____

Game: Around the World

Name: _____ Date: _____

Play with a parent or sibling. One person is the 'champion' and stands. Read a fact; first correct answer advances!

How to play: Reader says a fact (e.g. "7 times 8"). First to say the product correctly moves to the next seat. If the champion is beaten, the winner becomes the new champion. Go all the way around!

Fact list (reader keeps this page):

1. 5 × 6	2. 0 × 9	3. 8 × 10	4. 8 × 9
5. 3 × 6	6. 9 × 12	7. 2 × 8	8. 3 × 9
9. 5 × 12	10. 5 × 9	11. 5 × 8	12. 6 × 10
13. 5 × 8	14. 8 × 11	15. 2 × 7	16. 4 × 7
17. 10 × 12	18. 8 × 8	19. 4 × 7	20. 7 × 11
21. 1 × 7	22. 3 × 8	23. 1 × 9	24. 11 × 12

Game: Product Bingo

Caller picks a fact from the list. Mark the PRODUCT on your board. Five in a row (any direction) wins!

49	96	100	66	88
99	56	18	22	32
24	9	FREE	72	64
84	35	77	70	55
60	20	120	80	28

Game: Memory Match Cards

Name: _____ Date: _____

Print this page. Cut out the cards. Mix them face-down. Flip two: a fact and its product make a match!

Fact cards (cut these out):

6 × 7	6 × 8	6 × 9	7 × 7	7 × 8	7 × 9
8 × 8	8 × 9	9 × 9	8 × 12	7 × 12	12 × 9

Product cards (cut these out):

42	48	54	49	56	63
64	72	81	96	84	108

How to play Memory

1. Cut out all 24 cards. Mix them and place face-down in a grid.
2. Players take turns flipping two cards. A fact + matching product = keep the pair!
3. If they don't match, flip them back. Remember where they were.
4. Player with the most pairs at the end wins. Play again with a timer for extra challenge!

For Grown-Ups — Tips & Answer Keys

Short, happy practice beats long drills. 5–10 minutes a day is plenty.

Daily practice tips

- Warm up with a known strategy (9s finger trick, doubles) before mixed drills.
- Celebrate effort: "You used the turn-around!" matters more than speed at first.
- Focus hard facts (6,7,8,9,12) 3–4 times a week; easier facts for confidence boosters.
- Pair the printable with the offline HTML game (Practice by table → Race → Missing factor).
- Keep sessions short. Stop while it's still fun. Revisit missed facts the next day.

Drill A answers

9×12=108	9×9=81	0×4=0	5×7=35	6×10=60
1×9=9	3×3=9	1×3=3	3×8=24	5×8=40
7×10=70	2×4=8	10×12=120	7×9=63	8×10=80
0×0=0	8×8=64	3×10=30	0×7=0	7×7=49
8×12=96	2×9=18	0×8=0	6×12=72	1×8=8
3×9=27	1×7=7	6×8=48	4×5=20	5×6=30
1×6=6	0×10=0	6×7=42	3×6=18	2×12=24
2×8=16	10×7=70	12×11=132	9×4=36	6×2=12

Drill B (missing factor) answers

10×__=110 → 11	__×4=20 → 5	__×8=64 → 8
12×__=144 → 12	12×__=60 → 5	10×__=120 → 12
__×2=14 → 7	12×__=24 → 2	3×__=24 → 8
2×__=24 → 12	__×5=35 → 7	__×10=120 → 12
__×11=66 → 6	__×9=54 → 6	3×__=36 → 12
10×__=30 → 3	11×__=88 → 8	6×__=60 → 10
3×__=27 → 9	__×12=84 → 7	8×__=88 → 11
2×__=20 → 10	11×__=33 → 3	__×11=77 → 7
__×3=12 → 4	__×2=12 → 6	6×__=18 → 3
9×__=72 → 8		

Word problems + related facts

P1: 56	P2: 54	P3: 48
P4: 72	P5: 84	P6: 8
6×8=48 & 8×6=48	7×9=63 & 9×7=63	8×12=96 & 12×8=96
9×6=54 & 6×9=54		

Answer Keys (continued)

Timed challenge, Around the World, Memory Match, and fact families.

Timed challenge answers

1. 5×8=40	2. 5×11=55	3. 7×12=84	4. 7×7=49
5. 1×8=8	6. 4×12=48	7. 3×8=24	8. 6×12=72
9. 0×9=0	10. 1×7=7	11. 3×5=15	12. 6×6=36
13. 3×11=33	14. 5×12=60	15. 1×6=6	16. 2×9=18
17. 2×6=12	18. 4×10=40	19. 1×9=9	20. 4×9=36
21. 0×12=0	22. 3×7=21	23. 3×6=18	24. 3×12=36
25. 6×10=60	26. 2×10=20	27. 8×11=88	28. 9×9=81
29. 1×3=3	30. 4×8=32	31. 4×6=24	32. 0×2=0
33. 6×7=42	34. 0×7=0	35. 0×6=0	36. 0×11=0
37. 9×12=108	38. 0×3=0	39. 4×11=44	40. 7×8=56

Around the World products (for the reader)

5×6=30	0×9=0	8×10=80	8×9=72
3×6=18	9×12=108	2×8=16	3×9=27
5×12=60	5×9=45	5×8=40	6×10=60
5×8=40	8×11=88	2×7=14	4×7=28
10×12=120	8×8=64	4×7=28	7×11=77
1×7=7	3×8=24	1×9=9	11×12=132

Memory Match pairs

6×7 → 42	6×8 → 48	6×9 → 54	7×7 → 49
7×8 → 56	7×9 → 63	8×8 → 64	8×9 → 72
9×9 → 81	8×12 → 96	7×12 → 84	12×9 → 108

Fact-family triangles (page 2)

Each blank circle is the missing factor or product.
From every family write two multiplications and two divisions (turn-arounds count!).
Example: 6, 7, 42 → 6×7=42, 7×6=42, 42÷6=7, 42÷7=6.